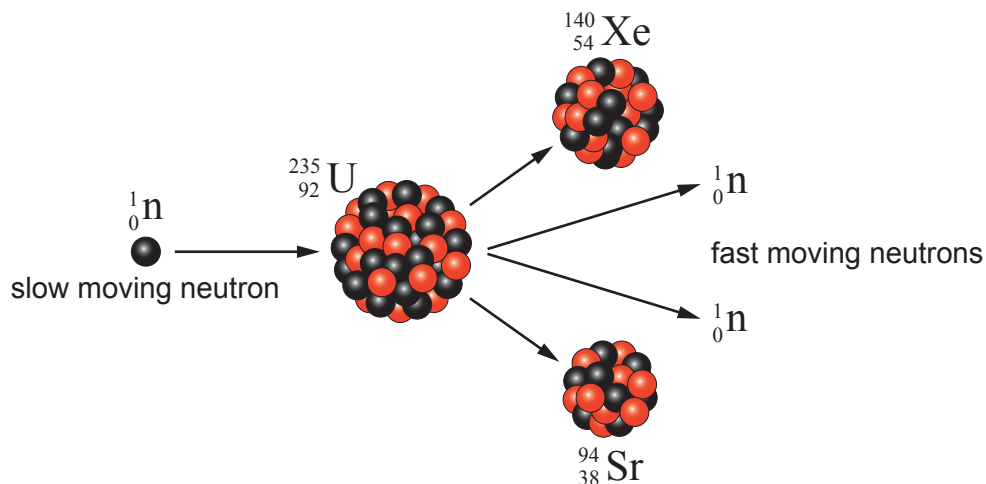


7. The diagram shows the fission of a uranium (U) nucleus by a slow-moving neutron (${}^1_0\text{n}$). This reaction can take place in a nuclear reactor. The fission produces nuclei of xenon (Xe) and strontium (Sr) and further neutrons.



- (a) Explain the function of the moderator in a nuclear reactor. [2]

.....

.....

.....

- (b) **Complete the nuclear equation** for the reaction shown in the diagram above. [3]



- (c) In the nuclear reaction shown above, only one of the neutrons causes further fission.

Underline the correct word from each bracket below to complete the sentences correctly. [2]

The other neutron is (**emitted** / **absorbed** / **reflected**).

This is done by (**control** / **fuel** / **moderator**) rods.

